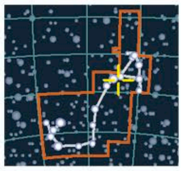
NEUTRON STAR

PSR B1620-26



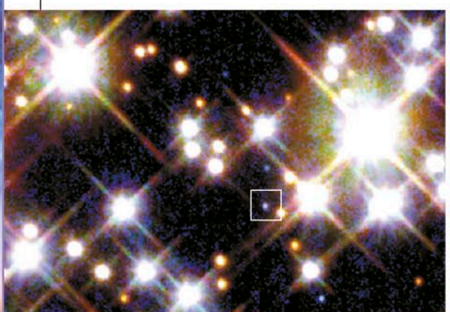
CATALOG NUMBER PSR B1620-26

DISTANCE FROM SUN 7,000 light-years

MAGNITUDE 21.3

SCORPIUS

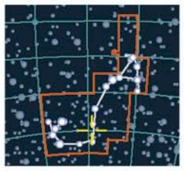
Situated in the globular cluster M4, the pulsar PSR B1620-26 rotates more than 90 times per second and has a mass of about 1.3 solar masses. It has a white-dwarf companion (boxed in the image below). A third companion is thought to be a planet twice the mass of Jupiter (see pp.296–299). This planet is named Methuselah, as it may be up to 13 billion years old.



WHITE-DWARF COMPANION

BLACK HOLE

GRO J1655-40



CATALOG NUMBER V* V1033 Sco

DISTANCE FROM SUN 6,000–9,000 light-years

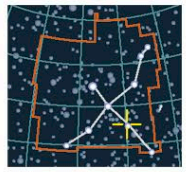
MAGNITUDE 17

SCORPIUS

Discovered in 1994, as a source of unusual X-ray emissions, this black hole produces outbursts in which jets of material are ejected at speeds close to the speed of light. In addition to this, the gas surrounding GRO J1655-40 displays an unusual flicker (at a rate of 450 times per second) that can be explained as a rapidly rotating black hole. This is only the second object of this type to have been found in the Milky Way. It has been suggested that a subgiant star is orbiting the black hole, which is six to seven times the mass of the Sun. Their orbits are thought to be inclined at 70 degrees to each other, causing partial eclipses. Mass has been pulled off the subgiant star by the gravitational interaction from the black hole and formed a disk of material around the system. This system has been dubbed a mini-quasar because of its similarity to active galactic nuclei (AGNs) (see pp.306–309).

BLACK HOLE

Cygnus X-1



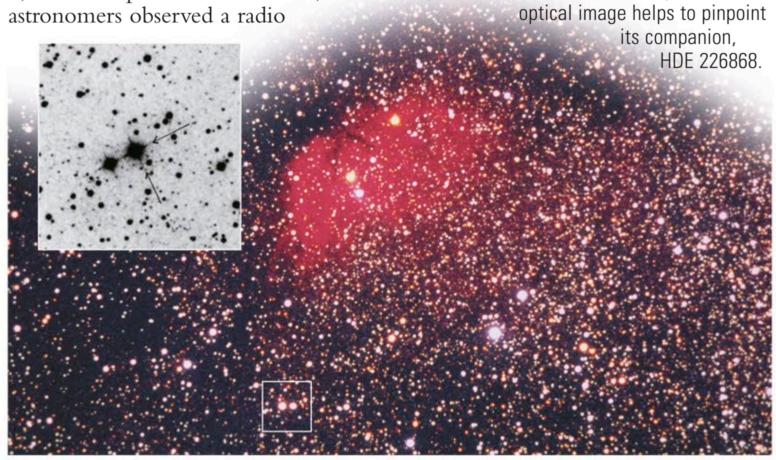
CATALOG NUMBER HDE 226868

DISTANCE FROM SUN 8,200 light-years

MAGNITUDE 8.95

CYGNUS

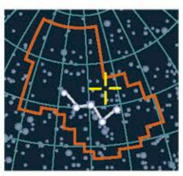
This X-ray source was one of the first to be discovered and is one of the strongest X-ray sources in the sky. The X-ray emissions from Cygnus X-1 flicker at a rate of 1,000 times per second. In 1971, astronomers observed a radio source at the same position in the sky and also identified an optical object, the blue supergiant star HDE 226868. This star has a mass of 20–30 solar masses and is visible through binoculars. It is in a 5.6-day orbit with Cygnus X-1, which has a mass of about six solar masses. Further observations have shown that the black hole is slowly pulling material from its companion supergiant and increasing its own mass. Cygnus X-1 was the first object to be identified as a stellar-mass black hole.



ELUSIVE BLACK HOLE
Cygnus X-1 is located close to the red emission nebula Sh2-101, within the rich Cygnus Star Cloud (below). A negative optical image helps to pinpoint its companion, HDE 226868.

SUPERNOVA

Tycho's Supernova



CATALOG NUMBER SN 1572

DISTANCE FROM SUN 7,500 light-years

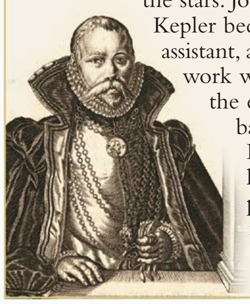
MAXIMUM MAGNITUDE –3.5

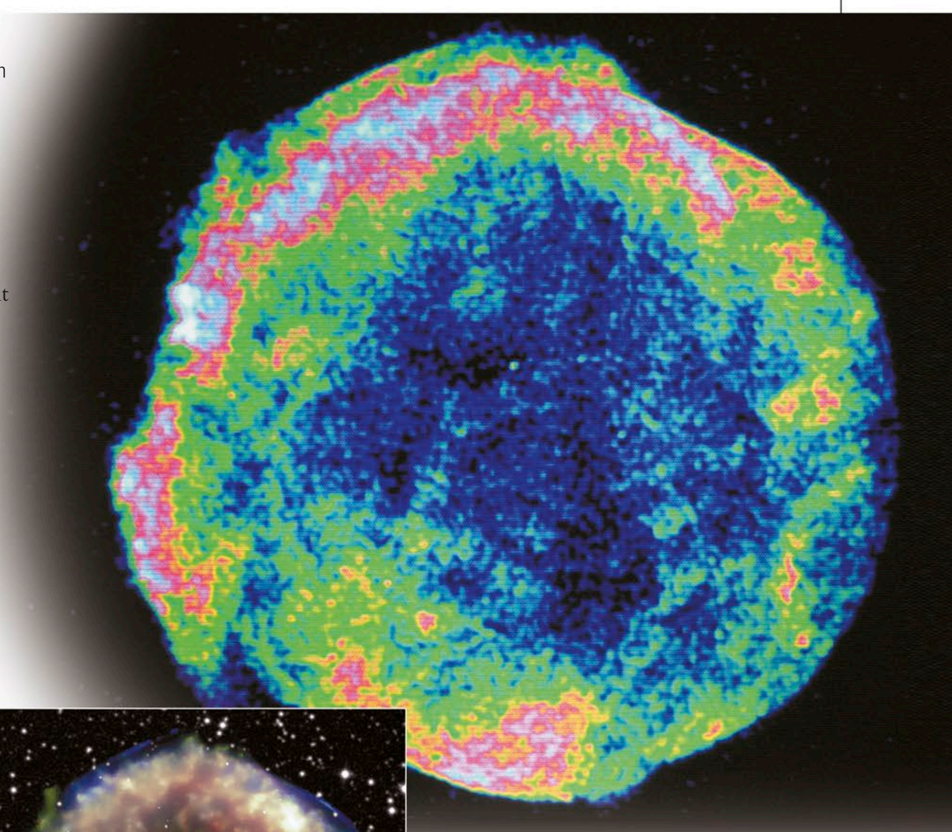
CASSIOPEIA

In 1572, Tycho Brahe (see panel, below) observed a supernova in the constellation Cassiopeia and recorded its brightness changes in exceptional detail. It brightened to around –3.5—as bright in the sky as Venus—before fading over a period of 6 months. This brilliant new object was to help astronomers reject the idea that the heavens were immutable. The remnant from this supernova is still expanding and has a current diameter estimated at nearly 20 light-years. Its stellar material is estimated to be traveling at 14.5–18 million mph (21.5–27 million kph), which is the highest expansion rate observed for any supernova remnant. No strong central point source is detected in the remnant, which suggests that Tycho was a Type Ia supernova. The model for this type of supernova is the destruction of a white dwarf when infalling matter from a companion star increases its mass beyond the Chandrasekhar limit (see pp.266–267). This concurs with the recent discovery of what astronomers think is the burned-out star from the heart of the supernova. The star was discovered because it is moving at three times the speed of other objects in the region. At the edge of the remnant is a shock wave heating the stellar material to 36 million °F (20 million °C); the interior gas is much cooler, at 18 million °F (10 million °C).

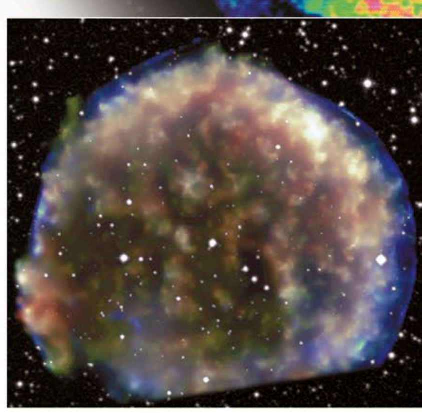
TYCHO BRAHE

The leading astronomer of his day, Tycho Brahe (1546–1601) founded a great observatory in Uraniborg, Denmark, and spent years making detailed observations of planetary movements and the positions of the stars. Johannes Kepler became his assistant, and Tycho's work was to give the empirical basis for Kepler's laws of planetary motion (see p.68).





RADIO ENERGY
A radio image of Tycho's Supernova shows areas of low (blue), medium (green), and high (red) energy. A shock wave produced by the expanding debris is shown by the pale blue circular arcs on the outer rim.



DEBRIS CLOUD
A Chandra Telescope X-ray image shows a false-color, wide-field view of the region around Tycho's Supernova. The image is cut off at the bottom because the southernmost region of the remnant fell outside the field of view of the Chandra camera.